北京邮电大学 本科毕业设计（论文）任务书

Project Specification Form

Part 1 - Supervisor

论文题目 Project Title	Network Resource Management Based on Language Model Multi-Agent Systems		
题目分类 Scope	Research	Networks and Wireless	Software
主要内容 Project description	This project explores network resource management based on language model multi-agent systems, aiming to address the growing complexity and dynamic demands of modern communication networks. With the rapid expansion of cloud services, data-intensive applications, and heterogeneous infrastructures, efficient resource allocation has become a critical challenge. Traditional centralized management approaches often struggle with scalability, adaptability, and real-time responsiveness. To overcome these limitations, this project leverages large language models (LLMs) integrated with multi-agent systems to enable distributed, intelligent decision-making. Each agent, powered by an LLM, is capable of interpreting high-level requirements, reasoning about network conditions, and collaboratively negotiating resource usage. The methods involve task decomposition, semantic reasoning, and reinforcement learning-based optimization for adaptive scheduling and load balancing. Tools such as Python, simulation frameworks (e.g., ns-3), and machine learning libraries (e.g., PyTorch) will be employed. By combining natural language understanding, multi-agent coordination, and algorithmic optimization, the project aims to deliver a scalable, resource-efficient, and intelligent management framework for next-generation communication networks.		
关键词 Keywords	Network Resource Managemen, Language Model, Multi-Agent Systems		
主要任务 Main tasks	1 Requirement Analysis and Modeling Identify the key challenges in network resource management and model them as tasks suitable for language model multi-agent systems.		
	2 Design and Development of Multi-Agent Framework Build a framework where language model agents can interpret network requirements, interact, and coordinate resource allocation decisions.		
	3 Implementation of Algorithms and Tools Apply optimization techniques, reinforcement learning, and semantic reasoning within the multi-agent system using tools such as Python, PyTorch, and network simulators.		
	4 Evaluation and Performance Validation Test the framework in simulated network environments (e.g., ns-3) to measure scalability, efficiency, and adaptability, and compare results against traditional approaches.		
主要成果 Measurable outcomes	1 Practical Skills in AI and Networking Students will gain hands-on experience with large language models, multi-agent system design, and network simulation tools, building interdisciplinary expertise in AI-driven network management.		
	2 Ability to Design and Evaluate Intelligent Systems Students will learn how to design algorithms for resource allocation, implement optimization and		

	reinforcement learning methods, and critically evaluate system performance in complex environments.
	3 The multi-agent demo system for resouce allocation simulation.